

Data Flow Diagram

Date	29 September 2026
Team ID	SWTID-2026-7019
Project Name	EduGenie - AI Learning Assistant
Maximum Marks	2 Marks

Data Flow Diagram

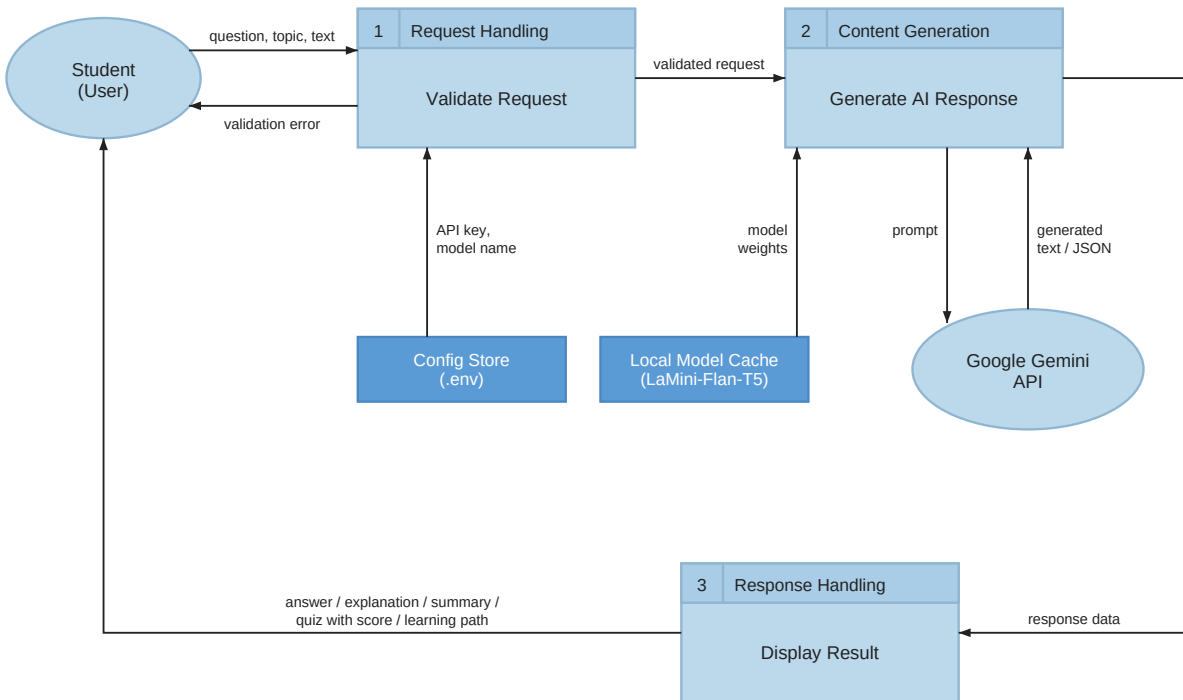
Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

EduGenie Data Flow Diagram

The diagram on the next page shows how a student's request moves through EduGenie - from the browser, through FastAPI validation, to the AI models (Google Gemini API and the local LaMini-Flan-T5 model), and back to the screen.



Element	Description
1 Request Handling	FastAPI route receives the question, topic or text from the browser, rejects empty input (HTTP 400) and checks that the Gemini API key is configured.
2 Content Generation	Builds the prompt for Q&A, explanation, summary, quiz or learning path; calls Gemini (or the local model for explanations) and parses quiz output into JSON.
3 Response Handling	Browser JavaScript renders the result, scores the quiz and shows errors as messages.
Config Store (.env)	Holds GEMINI_API_KEY, GEMINI_MODEL and LOCAL_EXPLANATION_MODEL.
Local Model Cache	Downloaded LaMini-Flan-T5-783M weights used for on-device explanations.